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MODEL EVALUATION[1]

Overview and Purpose

In today's data-driven organizations, deploying a model without a rigorous evaluation plan is a managerial risk. The quality of your metrics and tuning choices directly affects budgets, customer experience, and strategic actions. This technical note aims to connect the technical rigor of model evaluation and improvement with the managerial discernment required to decide when a model is 'good enough' for decision-making.

This note assumes the reader has knowledge of regression and classification models, and potentially also of tree-based models (including Random Forests). From that baseline, this note equips you to (1) judge model performance with the right metrics and (2) avoid common pitfalls like overfitting and leakage.

The Role of Evaluation in Data-Driven Decision Making

Since performance metrics translate technical outputs into business impact, poorly chosen metrics can optimize the wrong outcome. For example, misclassifying loyal customers as "at risk" inflates retention spending through unnecessary discounts, while missing true churners erodes revenue

[1] This work was developed with the assistance of artificial intelligence tools for content review, analysis, and editing tasks. All final decisions and authorship of the text correspond to the author.

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through losing clients. There is no universally "right" metric or threshold for each application, as each situation's "right" metric and threshold depend on which type of error carries higher cost in that business context.

It is important to mention that evaluation in a data-driven organization is not just about obtaining the highest scores or getting perfect predictions on our models; it is about understanding the business and knowing what the company's actual risk appetite is. As such, the bridge between building models and making business decisions is about translating how willing the company is to take on uncertainty, how much instability it can handle, and which risks could translate into concrete P&L and service results. A score simply is not applicable in this case.

The purpose should be a solid, useful evaluation: How the marginal value of an intervention changes across different segments; how that value diminishes over time. We also need to create a common language so that once things are deployed, executive decisions still make sense down the road.

Moreover, doing methodical, rigorous evaluation is not negotiable. It does not mean chasing the highest possible score, but that being disciplined about the assumptions made about the objectives, being honest about our data limitations, properly quantifying uncertainty based on actual business outcomes, and running sensitivity analyses.

In practice, a smaller but well-designed evaluation with clear tolerances, calibrated estimates, and reproducible evidence will generate better dashboards and, as a result, improve decision-making process quality.

Training, Validation, and Test Sets

Good practice in model evaluation suggests splitting data into three sets: training, validation, and test. Each data split serves a distinct purpose. The training set teaches the model patterns; the validation set helps choose hyperparameters and thresholds; and the test set provides a final, unbiased estimate of performance.[2]

A simple hold-out split might look like this:

$$[Raw\ Data] \rightarrow [Train\ 60-80\%] \ \ | \ \ [Validation\ 10-20\%] \ \ | \ \ [Test\ 10-20\%]$$

A common machine learning technique for determining these sets is cross-validation (CV), that rotates multiple folds to reduce variance in performance estimates, rather than relying on one validation slice. Stratified CV should be used for classification tasks to preserve class ratios, while time-series CV (with rolling or expanding windows) is preferred when temporal order matters. When using cross-validation, the validation step occurs internally within each training fold, so a separate validation set may not be required—but the test set should always remain untouched for final evaluation.

Key Phenomena to Monitor After Data Split and Model Training

Once the datasets are split and the model has run, it is crucial to keep an eye on certain phenomena that can seriously affect the evaluation results. Two particularly important ones are *overfitting* and *data leakage*:

[2] A simplified version is splitting data into a training and a testing set. Although the intuition of testing in data not used for the model's training remains, it runs into the risk of tuning the model for the testing set.

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- Overfitting occurs when a model performs excellently on the training data but poorly on validation or test sets, since it has learned noise. Underfitting, on the other hand, results when a model is too simple to capture underlying patterns and performs poorly on all sets.
- Data leakage is one of the most critical pitfalls. If information from validation or test data leaks into the training process, such as fitting a scaler on the full dataset or using the target variable during feature creation, metrics will appear overly optimistic. Always fit preprocessing steps inside the training fold or pipeline only.

Evaluation Metrics

Choose metrics that reflect business costs, not just mathematical elegance. In other words, a simple metric that can be easily understood by an executive audience and used to make decisions is often better than a sophisticated metric that is more appealing for mathematics or statisticians.

Classification (Binary or Multiclass)

In classification, the most used tool is the confusion matrix, with the corresponding metrics listed below.

Confusion Matrix Terms

Symbol	Full Name	Meaning	Example (Churn Prediction)
TP	True Positive	The model correctly predicts the positive class.	Predicted "will churn (1)" → actually churned.
FP	False Positive	The model incorrectly predicts the positive class.	Predicted "will churn (1)" → actually stayed.
FN	False Negative	The model incorrectly predicts the negative class.	Predicted "will stay (0)" → actually churned.
TN	True Negative	The model correctly predicts the negative class.	Predicted "will stay (0)" → actually stayed.

Note: As a tip to aid concept learning, "Positive" or "Negative" refer to the predicted classes, while "true" or "false" indicate whether that prediction was correct.

Core Metrics

- Accuracy = $\frac{\mathrm{TP+TN}}{(\mathrm{TP+FP+FN+TN})}$ ("Of predicted values, how many were right?")
- Precision = $\frac{TP}{(TP+FP)}$ ("Of predicted positives, how many were right?")
- Recall (Sensitivity) = $\frac{TP}{(TP+FN)}$ ("Of actual positives, how many did we catch?")
- F1-Score = $2 \times \left(\frac{(Precision \times Recall)}{(Precision + Recall)}\right)$ (balances P/R)
- ROC & AUC: Ranking quality across thresholds; may be optimistic on imbalanced data. The higher the AUC, the better the model.
- PR Curve & Average Precision: Focuses on precision–recall; preferred for rare positives. Presents the balance between improvement in precision or recall.

In business terms, whether Accuracy, Precision or Recall (Sensitivity) matter more depends on the application. Accuracy is very intuitive and adequate with class balance, which means having a similar number of positives and negatives; for instance, head or tail throws of a coin. However, most business applications have class imbalances. Class imbalance is having a different proportion of positives and negatives; for instance, 95 non-defaulters for 5 defaulters in a loan. A prediction of all 100 loans not defaulting would have a high accuracy but would not be very useful for making decisions. In those cases, Precision and Recall are more relevant.

Consider a fraud or loan default example. If missing a defaulter (a false negative) is costlier than wrongly flagging a good borrower (a false positive), it is better to optimize for higher recall. In contrast, if wrongfully flagging a good borrower (a false positive) is costlier than missing a defaulter (a false negative), then it is better to optimize for higher precision. Presenting cost-based trade-offs to executives helps align model optimization with financial goals.

Regression

Regression models are evaluated using measures of prediction error. Mean Absolute Error (MAE) expresses the average absolute deviation between predicted and actual values and is interpretable in original units, making it robust to outliers. Mean Squared Error (MSE) penalizes larger errors, while Root Mean Squared Error (RMSE) (the square root of MSE) highlights large deviations and shares the same units as the target variable. The coefficient of determination, R^2, measures the proportion of variance explained by the model but should be used carefully when assumptions are violated. Mean Absolute Percentage Error (MAPE) provides a percentage-based interpretation but fails near zero values and over-penalizes small denominators.

Core Metrics Calculations

- MAE (mean absolute error) = $\frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$, interpretable in original units, robust to outliers.[3]
- MAPE (mean absolute percentage error) = $\frac{1}{n}\sum_{i=1}^{n}\frac{|y_i-\hat{y}_i|}{y_i}$ can be intuitive but fails near zero and over-penalizes small denominators.
- MSE (mean squared error) = $\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$, penalizes large errors.
- RMSE (root of the MSE) = $\sqrt{MSE}$: same units as target; highlights large errors.
- $R^2 = 1 - \left(\frac{SSE}{SST}\right)$: variance explained; use with care when assumptions break.

[3] **Note:** For those unfamiliar with mathematical terminology, these formulas might look daunting. They are actually easy to follow and their interpretation can help you understand the concepts. Here we describe the MAE formula step by step:

i. $\frac{1}{n}$ means the final value will be divided by *n*, which is the number of observations in the dataset.

ii. $\sum_{i=1}^{n}$ means a sum from the first observation (i=1) to the last one (i=n) will be done. Together, $\frac{1}{n}\sum_{i=1}^{n}$ implies we are computing a **mean.**

iii. $|y_i - \hat{y}_i|$, a value between | | means it is an **absolute** value – that is, regardless of whether it was positive or negative, we take it as negative. Both 2 and -2 have an absolute value of 2, for example. y_i stands for the actual value of observation *i.* $\hat{y}_i$ *(y hat) stands for the predicted value of observation i. Together, they mean* ***the absolute error*** *(the absolute difference between the actual and the predicted value).*

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Similarly to classification, the best metric depends on the business application. In a monthly sales forecasting context, RMSE is more informative if large misses disrupt operations (like stockouts), whereas MAE might serve better for budgeting where average absolute deviations matter more.

Managerial Insights

- Therefore, "best" equals fit for purpose, not "most complex". Tie your metric to business KPI and prioritize explainability and usefulness for making decisions over technical complexity.
- Always validate results on hold-out data, and when feasible, confirm robustness through a time-based back test or controlled A/B test before scaling.
- Watch class imbalance. Report PR-AUC, use stratified CV, consider class weights or resampling (carefully, inside CV only).
- Communicate with thresholds, not only scores. Show how moving the decision cut-off shifts precision, recall, and cost.
- Choose metrics carefully, justifying with the business application which one is more relevant. When it is unclear which one is best, compare multiple metrics and analyze the insights of each.
- Model evaluation should be seen as a learning process that aligns model improvements with measurable business outcomes.

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EXHIBIT 1
Quick Reference: Classification Metrics

Metric	What it captures	Strengths	Watch-outs	Use when…
Accuracy	Overall correctness	Simple, intuitive	Misleading with imbalance	Classes balanced, equal error costs
Precision (P)	Purity of positives	Limits false alarms	Can reduce recall	False positives are more costly
Recall (R)	Coverage of positives	Catches more true cases	Can reduce precision	False negatives are more costly
F1-Score	Balance of P/R	Single number	Hides class nuance	Both P and R matter
ROC-AUC	Ranking ability	Threshold-independent	Optimistic in imbalance	Early screening, broad ranking
PR-AUC	Precision vs. recall	Focus on positives	Less intuitive	Rare-event problems

EXHIBIT 2
Quick reference: Regression Metrics

Metric	What it captures	Strengths	Watch-outs	Use when…
MAE	Average absolute error	Interpretable, robust	Treats all errors equally	Budget variance, median-like robustness
RMSE	Large-error sensitivity	Penalizes big mistakes	Sensitive to outliers	SLAs/logistics where big misses hurt
R^2	Variance explained	Scale-free	Can be inflated	Exploratory fit on stable variance

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